Answering a busy group chat without a backlog

A design for an agent that works on many chat requests at once, still gives every reader a legible stream of answers, and survives people changing their minds while it is already working.

Jay Talwar Written before the code 18 October 2026

Before you start

This document assumes no background. Every technical term is defined where it first appears, and there is a glossary at the end. If you hit a word you don't recognise, it is defined somewhere above — or it shouldn't be there.

Here is the whole world the design lives in:

- An **agent** is a program that answers questions using a language model. To answer "am I overweight energy?" it has to *look things up* — fetch your holdings, fetch prices. Those lookups are called **tool calls**, and they are the slow, expensive part. A single question might take twenty seconds and cost a few cents.
- Cost is measured in **tokens** — roughly, pieces of words. Everything the agent reads and writes is billed by the token. When this document says work was "wasted", it means tokens we paid for and threw away.
- The agent sits in a **group chat** with several people in it. They talk at the same time, interrupt, change their minds, and ask follow-ups.

That is it. Everything else is built up from here.

§ 01

The problem, concretely

Castle's agent sits in a group chat. When the chat gets busy, several people ask it several things at once. Today it answers them one at a time, so a queue forms. The brief asks for a system that works on many requests **in parallel** — several at the same time, rather than finishing one before starting the next — still gives each chat a **clear, ordered stream of answers**, and handles people **changing or adding to a request while it is running**.

Here is the situation the system has to survive. Times are seconds from the start of the burst.

t=0	**Ana**	What's my exposure to a Taiwan escalation?	NEW · SLOW
t=2	**Ben**	Is NVDA still my biggest position?	NEW · FAST
t=4	**Ana**	Actually make that Taiwan *or* South China Sea.	CHANGES A RUNNING REQUEST
t=5	**Cara**	What's my exposure to Taiwan?	SAME QUESTION ANA ASKED
t=9	**Ben**	And the second biggest?	NEEDS AN EARLIER ANSWER
t=11	**Cara**	never mind the Taiwan one	CANCELS

Six messages, three pieces of work. Two of the three change shape after they have already started. This transcript is the specification — and scenario 1 of the test suite.

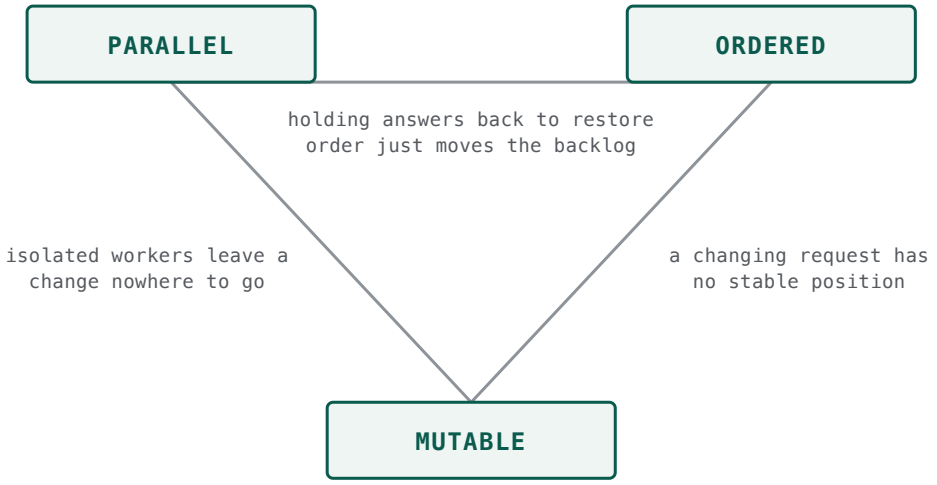
> Throughout this document, a **`request`** means one piece of work the agent owes somebody an answer for. A **`message`** is one line somebody typed. The two are not the same thing, and § 04 argues that confusing them is the central mistake available here.

A system that treats each message as an independent job gets this transcript wrong in at least four ways. It answers the Taiwan question twice — Ana and Cara asked the same thing. It answers Ana's original question, which she already replaced. It answers "the second biggest?" before anyone knows what the first one was. And it keeps paying for work Cara cancelled.

§ 02

Why the three requirements fight each other

The three requirements are not independently satisfiable. Every *pair* of them is in tension, which is why this is a design problem rather than a to-do list.



Each edge is a real conflict, not a tradeoff dial. The way through is to notice that one of the three corners is not what it appears to be.

Parallel × Ordered

Work finishes out of order — a question that takes one second finishes before one that takes thirty, whoever asked first. You could restore arrival order by holding finished answers back until their predecessors are done. But then a fast question asked third waits on a slow question asked first, and you have moved the backlog from *computing* answers to *delivering* them. The user sees no improvement.

> **Head-of-line blocking** — when the first item in a queue is slow, everything behind it waits, however fast those items are individually. It is the specific failure the brief describes as "answering them one at a time creates a backlog". Named after a queue of people at a door: one person fumbling for their ticket blocks everyone behind them.

Parallel × Mutable

A change — *"actually make that…"* — targets **one specific computation that is currently running**. The obvious way to get parallelism is to give each request its own isolated worker that shares nothing with the others. But then the change has nowhere to go: whoever receives the new message has no handle on the worker that needs to hear it, and no shared context in which to work out what "that" refers to.

Ordered × Mutable

"Ordered" presumes a stable sequence of things to order. If request #2 becomes a materially different request at t=30s — or is cancelled, or turns out to be the same question as request #4 — the sequence is not stable. Ordering by the arrival time of the *original* message is ordering by a fact that is no longer true.

§ 03

First reframe: "ordered" is about legibility, not sequence

Nobody in a group chat actually wants strict arrival order.

> **FIFO** — "first in, first out", the rule that things come out in the order they went in. A supermarket queue is FIFO. It is the obvious reading of "ordered", and it is the one that costs head-of-line blocking.

Pull on "a clear, ordered stream of answers" and ask what a reader actually needs, and it comes apart into four separate properties.

PROPERTY	WHAT IT MEANS FOR THE READER	NEEDS ORDERING?
Attribution	For every answer, I can tell which question it answers.	No
Completeness	I can tell what is still outstanding. Nothing silently vanished.	No
Non-contradiction	I never see two live answers that disagree. A replaced answer is visibly replaced.	No
Causal consistency	If answer B depends on answer A, A arrives first.	**Yes** — and only between dependent pairs

Three of the four need no ordering at all. FIFO delivery buys all four with one blunt instrument and charges head-of-line blocking for it.

Implemented directly, each property is cheap, and none of them blocks:

- **Attribution** → quote the question back: `re: "exposure to Taiwan" — …`
- **Completeness** → acknowledge each request when it arrives, and keep the outstanding set visible

- **Non-contradiction** → say “this replaces my earlier answer” when it does, and never ship an answer belonging to an out-of-date version of a request (§ 07)
- **Causal consistency** → record which request depends on which, and serialize *only* across that link

> THE DESIGN THESIS
>
> **Serialize on dependency, not on arrival.**

Serialize here means “force to happen one after another”. The claim is that the only thing that genuinely has to wait is an answer that needs another answer first — and nothing else.

Everything downstream follows from that sentence. It turns ordering from a structural property of the system into a **policy** that can be swapped per chat — which in turn makes it something we can *measure* rather than assert (§ 09, § 10).

§ 04

Second reframe: messages are not requests

The brief’s phrasing — “several questions at once” — invites you to treat each message as a job. In a real group chat that is wrong. Look at the transcript again: six messages, three requests.

So the right question on the way in is not *“what does this message ask for?”* but **“how does this message relate to the work already in flight?”** That distinction is load-bearing. It is what lets “And the second biggest?” create the dependency link § 03 needs, and what lets Cara’s repeat of Ana’s question avoid paying twice for one answer.

The six relationships we handle

RELATIONSHIP	MEANING	WHERE TO SEE IT
NEW	An independent request.	transcript, Ana t=0 and Ben t=2
AMEND	Changes an existing request. Carries one flag, `preserves_premise`.	transcript, Ana t=4
CANCEL	Stop work; no answer is owed.	transcript, Cara t=11

RELATIONSHIP	MEANING	WHERE TO SEE IT
DUPLICATE	Someone already asked this; one answer serves both.	transcript, Cara t=5
REFERENCE	A new request that needs a previous request's *answer* first.	transcript, Ben t=9
IGNORE	Not a request at all.	`busy_chat` : "morning all", "thanks!!"

`IGNORE` is the odd one out, and it is listed because leaving it out would be dishonest about the classifier's job. The other five describe a relationship *between* this message and some existing request. `IGNORE` says there is no request here to relate to anything — it is a greeting, a thank-you, a joke, or one member answering another. The Sequencer's response is to return no commands at all, so `IGNORE` is the only relationship that never appears in the scheduling diagrams.

It still matters, for two reasons. **It is most of a real group chat.** A scheduler that treats every message as a request answers "morning all" with a portfolio report. And **it is where the dangerous mistakes live**: "wait" or "no" or "ok" are almost always social, but read as `CANCEL` they kill live work. Five of the labelled set's fourteen hard cases are exactly this — "interesting", "ok", "wait", "no", "is that still running?" — which is why § 10 reports the hard cases separately from the clear-cut ones.

The `preserves_premise` flag is doing real work. "Also include South China Sea" *widens* the question — the research already done on Taiwan is still valid, so we keep it. "Actually, not Taiwan, I meant inflation" *replaces* the premise — that research is now worthless and must be thrown away. Same relationship, one boolean, very different cost. Folding *widen*, *narrow* and *replace* into one class with one flag removes three categories the classifier would otherwise have to tell apart, and loses no behaviour.

Who writes the amended question

Classifying a message as `AMEND` is only half the job. Something still has to produce the question the worker is now meant to answer, and the obvious approach does not survive contact with real messages:

> original: *"How did my book do in Q2?"*
> change: *"sorry I meant Q3 not Q2"*
> glued together: *"How did my book do in Q2? sorry I meant Q3 not Q2"*

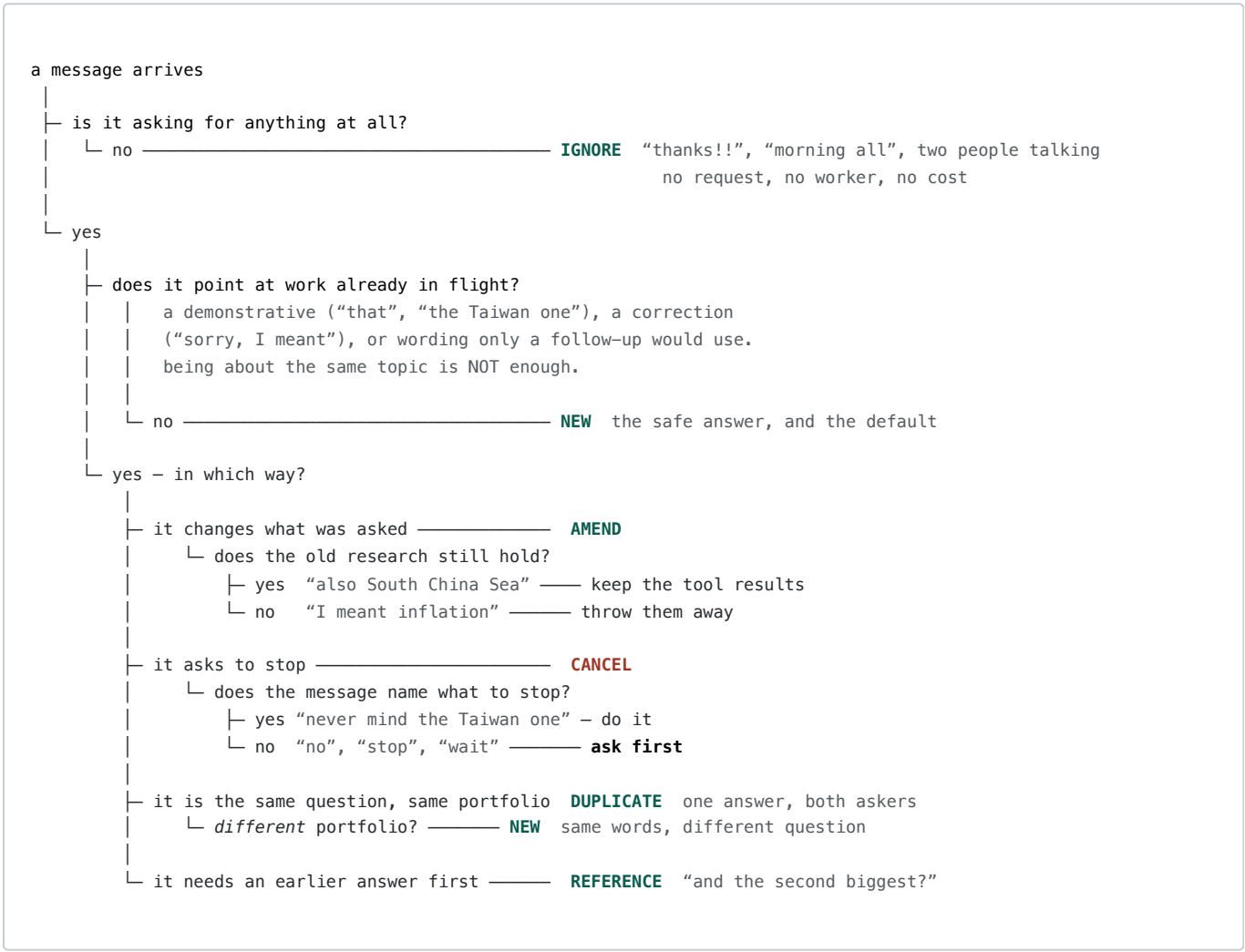
That hands the worker a self-contradiction and hopes it picks the right half.

So triage returns a third thing alongside the label and the flag: the merged question, already resolved — *"How did my book do in Q3?"*. The model has both texts in front of it and is the only component in a position to

combine them, so it is asked to state the result rather than leaving the Sequencer to concatenate strings it cannot read. The field is `rewritten_question` in the model's response and `rewritten_spec` on the event.

Concatenation survives as the fallback for when the model omits the field — glue on a widening change, replace outright on a premise-replacing one, picked by the same `preserves_premise` flag. It is a deliberately poor fallback, and the reason the Sequencer prefers the rewrite whenever one is offered.

This is also the clearest case of the division of labour in § 03: judgement about language lives in the model, and decisions about work live in the Sequencer. "What is this question now?" is a language question. "Whose slot does it get, and may the answer ship?" is not.



Everything hangs off two questions: is this a request, and does it point at something already running. Both default to the cautious answer, because the costs of being wrong are not symmetric.

What the classifier is shown

Triage sees two things: the **live request set** — every request still in flight, with its stage, asker, member and question — and the **recent chat**, bounded to the last six messages.

The second was missing for most of this project's life, and the gap was not obvious. The classifier's hardest job is telling a message aimed at the agent from one aimed at another person in the room, and the request set cannot settle that. Consider Ana typing:

```
ben: ana are you joining the 3pm call?
ana: no
```

`no` is an answer to Ben. Without the preceding line it is a bare word next to a live Taiwan request, and the honest reading is a coin flip — in five samples the classifier returned `IGNORE`, `NEW` *and* `CANCEL`, and `CANCEL` would have killed Ana's question. With the line, it answers `IGNORE` five times out of five.

> The eval set's own note on that case read *"almost certainly answering someone else in the chat"* — a label whose justification rested on evidence the classifier was never given. That is the kind of mistake that survives a long time, because every individual component looks reasonable.

	WITHOUT CHAT	WITH CHAT
Agreement with the labels	143/152 (94%)	**145/152 (95%)**
On the genuinely ambiguous cases	49/56	**52/56**
Spread across identical passes	92% – 97%	**95% – 97%**
IGNORE precision / recall	0.97 / 0.95	**1.00 / 1.00**
CANCEL precision / recall	1.00 / 0.95	**1.00 / 1.00**
REFERENCE precision / recall	0.89 / 1.00	**1.00 / 1.00**

The headline number barely moves, and it is the least interesting column. What changed is the shape.

The entire class of "two people talking to each other" errors is gone, `IGNORE` and `CANCEL` are exact, and the errors that remain are `NEW` → `AMEND` and `DUPLICATE` → `NEW` — under-merging, the recoverable direction.

The result that matters most is the third row. § 10 found the run-to-run variance large enough to swamp the confidence mechanism meant to control it. Supplying the missing evidence narrowed it, and two independent four-pass runs afterwards produced *identical* confusion matrices. **The instability was never really in the model; it was in asking a question that the available evidence could not answer.**

The window is capped at six messages for the reason § 08 gives: the live request set is bounded, so this must be too, or a chat that runs all day grows its own classification cost without limit.

Classifier errors must be asymmetric by design

> **Classifier** — a model whose job is to put a thing into one of a fixed set of categories. Here: given a message and the live request set, pick one of the six relationships above. **Confidence** is the classifier's own estimate of how likely it is to be right, from 0 to 1.

The classifier is a small language model and it will sometimes be wrong. What matters is making sure it is wrong in the **recoverable** direction.

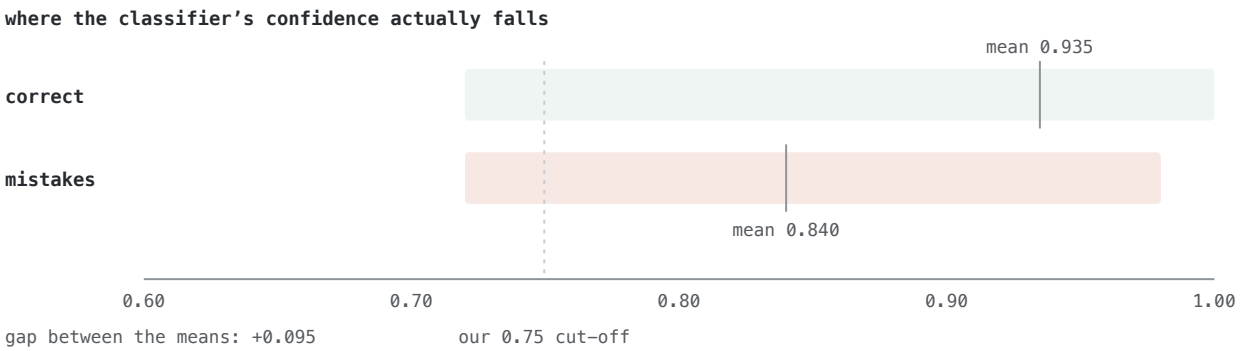
- A `NEW` misclassified as `AMEND` **destroys a request.** The original question is overwritten, never answered, and nobody finds out.
- An `AMEND` misclassified as `NEW` **produces a redundant answer.** Noisy, but every question still gets answered and the reader can see what happened.

The second failure is recoverable and the first is not. So the classifier is **biased toward** `NEW`: each relationship has a confidence threshold it must clear, the destructive ones (`CANCEL`, and `AMEND` that replaces the premise) have a higher bar than the additive ones, and anything below its threshold falls back to `NEW`.

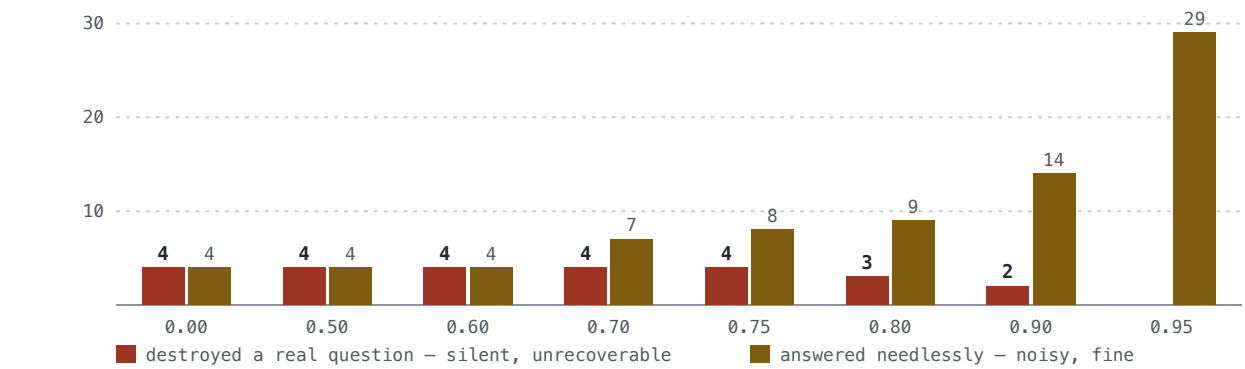
Because the classifier is the highest-risk component in the system, every decision it makes — proposed relationship, confidence, target, and whether a fallback was applied — is logged and printable. It is the one part a reviewer should be able to audit directly.

Measuring that, and finding it does not work

The argument above is sound. The thresholds implementing it — `0.75` and `0.50` — were invented, so § 10 measures them against a labelled set. The result is a negative one about this design.



The two bands are nearly the same band. A threshold works by sitting in a gap between right answers and wrong ones. There is no gap here.



The threshold buys little until it buys too much. From 0 to 0.90 the red bars move by two while the amber ones more than triple. Only at 0.95 do destroyed questions reach zero — by rejecting 29 predictions to save 4. One run; a second identical run started at 7 destroyed rather than 4.

The variance was the real finding. Two identical 152-case runs gave 7 destroyed questions and 4 — a spread as large as the effect the threshold was supposed to produce. That is what led to the chat context above. Supplying the missing evidence narrowed it to 4 on both runs. The threshold did not become useful (the bands still overlap), but the *reason* it looked useless turned out to be partly fixable, and fixing it was worth more than any tuning of the knob. **The instability was in the question, not the model.**

A threshold cannot buy this asymmetry, because legitimate amendments and mistaken ones arrive at the same confidence. Two things replace it.

A signal that is not confidence. Whether the message *names what it refers to* separates them cleanly. “Never mind the Taiwan one” names its target; a bare “no” — quite possibly aimed at another person in the chat — does not. So a `CANCEL` whose message names nothing **asks instead of acting**, and that costs no parked state: if they meant it, they say so again, and the reply names its target by construction. `CANCEL` precision is 1.00 with this in place.

Fixing the error the measurement actually found. The dominant failure was not over-cancelling but **over-merging** — reaching for `AMEND` and `DUPLICATE` where the honest answer was `NEW`. Telling the classifier about that specific bias, and that topical similarity is not evidence of relatedness, cut destroyed questions from 9 to 6 and `DUPLICATE` → `AMEND` errors from 4 to 1.

A signal that works, and why it is no longer needed

Showing that confidence fails raises an obvious question: is there *any* runtime signal that separates right answers from wrong ones? There is, and it is not the model’s opinion of itself — it is whether the model agrees

with itself. Classifying each message five times, unanimous answers were 97% correct and split ones 33–60%.

Two things made that more than a curiosity. **Majority voting did not help** — 3 errors against 3.4 for a single sample, and worse on a second run. The mistakes were systematic rather than random, so five samples reproduced the same bias five times. It was the *disagreement* that carried information, not the vote. And the messages it flagged were exactly the ones a person would hesitate over: “wait”, “no”, “is that still running?”.

Then the chat context above was added, and the measurement was repeated:

	BEFORE CHAT CONTEXT	AFTER
Cases where five samples disagree	5–6 of 38	**2 of 38**
Errors on a single sample	3.4–4.2	**1.8**
Destroyed questions	1.6–2.6	**1.0**
Correct when unanimous	97%	97%

The signal survived; the thing it was detecting mostly did not. Giving the classifier the evidence it was missing removed two thirds of the disagreements outright, because they were never really the model being unsure — they were the model being asked an unanswerable question and improvising a different answer each time.

> So it is not built, and the reason is better than the cost. Five triage calls per message would now buy a flag on two cases out of thirty-eight, one of which is genuinely ambiguous. **Detecting uncertainty is a worse fix than removing its cause**, and the measurement is what made the difference visible. If the remaining errors ever justify it, the mechanism is characterised and ready; it is simply not earning its keep.

The thresholds stay, but it is worth being exact about what they now do. They run on every classification, and across 114 measured classifications they fired **zero times** — mean confidence is 0.939 and the lowest seen is 0.72, while the destructive gate sits at 0.75 and only applies to `CANCEL` and `IGNORE`, which never arrived that unsure. So they are not “catching the low-confidence tail”; on this data there is no tail to catch. They are kept as a cheap backstop against inputs unlike the ones measured, and nothing more. The work is being done by `names_target` and by the prompt.

§ 05

Architecture

Four components. Only one of them holds state shared between requests, and it is deliberately the cheap one.

CHAT MESSAGES

several people, talking at once

every inbound message

TRIAGE

claude-haiku-4-5

Classifies each message against the live request set (§ 04) and emits one command. Runs on everything that arrives, so it must be cheap.

NEW / AMEND / CANCEL / DUPLICATE / REFERENCE / IGNORE

SEQUENCER — ONE PER CHAT

the only shared state · one event at a time · no model calls

- The register of live requests
- Which request depends on which
- **Admission** — what is allowed to start
- **Delivery** — what is allowed to ship

Pure bookkeeping.

start (request, version) → ← results, version-checked

WORKER POOL — BOUNDED

claude-opus-5

One isolated worker per request-version. Each has an inbox of pending changes that it reads only at safe moments.

answers

DELIVERY

Owns every reader-facing property from § 03: attribution, acknowledgements, replacement notices.

The shared-state component is the cheap one. Haiku runs once per inbound message; the Sequencer does no model calls at all; Opus runs only inside isolated, replicated workers.

Worker — one running job answering one request. **Pool** — the fixed set of workers that may run at once, say three. A request waits for a free **slot** before it starts. Bounding the pool is what stops a busy chat from launching forty expensive jobs at once.

Why not either sketch in the brief

Sketch 1 — a router plus parallel runtimes. A small model reads each message and sends it to its own isolated runtime. Good isolation, genuinely parallel. But the router is *stateless with respect to the live request set* — it does not know what is currently running. So it structurally cannot classify relationships: `AMEND`, `DUPLICATE` and `REFERENCE` are unrepresentable, because each of them is a statement *about an existing request*. A change arrives with no addressable target. And no component owns ordering, so requirement #2 has nowhere to live.

Sketch 2 — one runtime with subagents. The main agent answers easy things itself and spawns helpers for hard ones. Changes are natural here, because everything shares context. But the main runtime becomes a **serialization point**: every incoming change, every helper's result, and every delivery decision passes through one loop that is *also doing model inference*. Under a burst the backlog comes back — now bottlenecked on the one component you cannot replicate.

> THE ARCHITECTURAL CLAIM
>
> **Serialize the cheap thing. Parallelize the expensive thing.**

The brief's second sketch does the exact opposite: it serializes inference and parallelizes only the leaves. This is the central claim of the submission, so § 09 **measures** it rather than asserting it — there is a metric whose whole job is to falsify it.

Where this sits relative to the swarm post

The brief recommends Prime Intellect's *On the Nature of the Swarm* as background. It argues that hierarchical agent trees bottleneck at the root — the root's context fills no matter how large the swarm gets, recreating the original problem one level up — and that the way out is peer-to-peer meshes coordinating through shared state rather than through a coordinator.

It reaches the same verdict on the brief's second sketch, by a different route. The post's objection is that the root runs out of *context*; the objection above is that the root runs out of *throughput*, because it is doing inference on the critical path. Two different resources, the same structural fault: a component that cannot be replicated sits in the middle of everything.

It also names the escape route this design took. The post recommends coordinating through shared state — git, issue trackers, things agents read and write without anyone holding the whole picture. The Sequencer is exactly that: shared state that is deliberately *stupid*. It has no context to fill because it never reads an answer, and no throughput ceiling because it never calls a model. So a star topology with a dumb hub avoids the bottleneck the post identifies, without needing a mesh to do it.

Where it criticises this design, it was right — and it found a bug. The post argues for persistent agents over disposable ones, because a sub-agent that forgets everything on completion forces redundant work on the follow-up. Workers here are strictly disposable, one per request-version, no memory. Taking that seriously turned up something worse than waste.

A `REFERENCE` is an answer *about* an answer. But a follow-up admitted much later than its parent was being pinned to a **different data version** — so "and the second biggest?" could be computed against holdings the answer it explicitly builds on never saw, and the two could disagree. That is the same incoherence § 08 already fixes *within* a request, missed *across* a dependency edge. A follow-up now inherits its parent's data version and its tool results, which closes the coherence hole and removes the redundant lookup in the same move. Freshness is the wrong instinct here: if you want current data you ask a new question, not a follow-up.

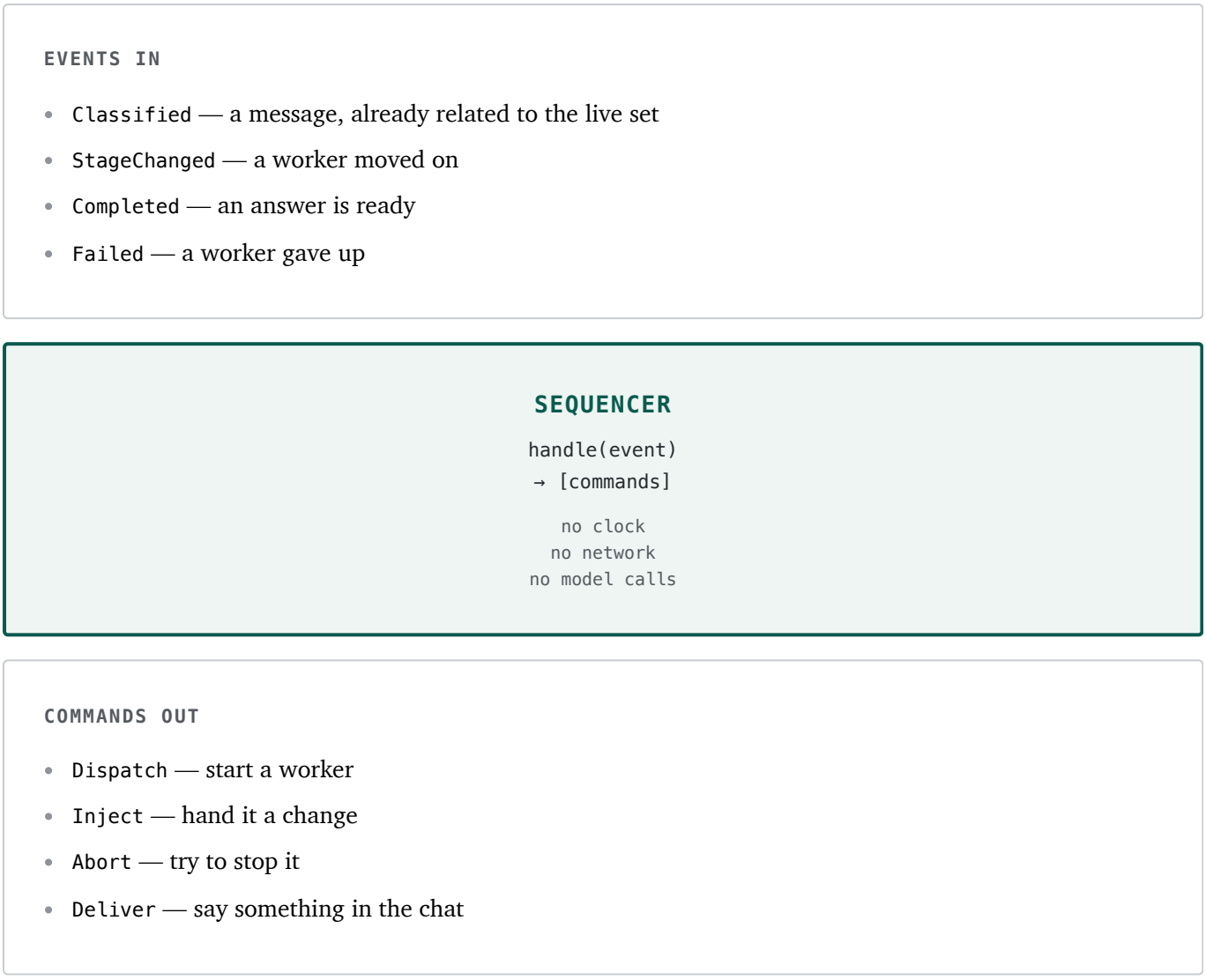
What remains unaddressed is the broader version of the same idea. The post observes that agents specialise naturally — whoever looked at the frontend first becomes the frontend expert — and that routing should exploit it. Inheritance only helps along a dependency edge; two *unrelated* questions about the same member at the same data version still each pay for the lookup. The general fix is a tool-result cache keyed by member, data version and call, which is a real design problem rather than a small one, and is not built.

One honest caveat on scale. The post's mesh argument is about swarms of ten thousand agents; this system runs a pool of three against one chat. At that size a dumb hub is not a compromise, it is the right answer — coordination costs nothing and a mesh would be ceremony. The argument starts to bite at a scale this design would have to grow into, and § 12 is honest that cross-chat scheduling is not built.

The Sequencer is a pure function

Pure function — code that takes input and returns output, touching nothing else: no files, no network, no clock. Given the same input it always returns the same thing.

The Sequencer is written as `handle(event) → [commands]`. It never calls a model, never reads the time (time arrives *on* each event), and never performs I/O. It returns a list of instructions — *start this worker*, *stop that one*, *ship this answer* — and something else carries them out.



It decides; something else acts. Time arrives *on* the event rather than being read, so the same function runs unchanged in the simulator and against real models — and a fuzzer can drive it thousands of times a second with no network.

This is not stylistic. It buys four concrete things:

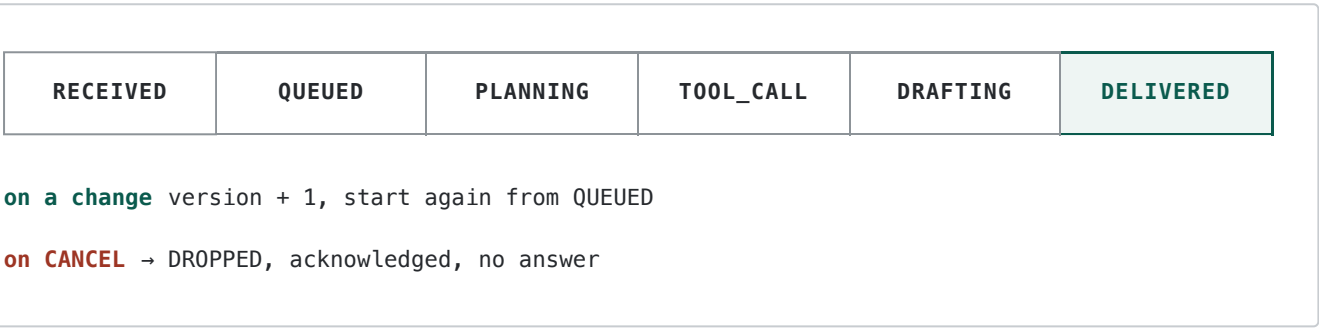
- **It makes the claim above structural.** "The Sequencer does no inference" stops being a promise and becomes something the shape of the function enforces.
- **The races in § 08 cannot happen.** One event is handled at a time, start to finish, so there is no moment where two things are half-applied.
- **It makes the testing in § 10 possible at all.** A pure function can be driven by a fuzzer thousands of times per second with no network calls.
- **Crash recovery comes free.** If the whole state is a fold over events — `state = fold(handle, events)` — then persisting the events *is* persisting the state. Recovery is replaying the log: no checkpoint format, no snapshot to keep in sync with the thing it describes, no reconciliation. § 14 was going to list this as future work; it turned out to be about eighty lines, which is the point.

The same Sequencer runs unchanged in the simulator and in live mode against real models. Only the thing driving it differs.

§ 06

How a request moves through the system

Every request carries a **version number** that goes up by one every time its specification changes. Every job started and every result returned is stamped with `(request, version)`. This one mechanism does most of the work in § 07.



QUEUED means waiting for a free worker slot. **PLANNING** is the model working out what it needs, **TOOL_CALL** is the expensive lookup, **DRAFTING** is writing the answer.

A queued request starts when **two** conditions hold: a worker slot is free, **and** everything it depends on has already been answered. The second condition is checked *before* a slot is taken, never while holding one — § 08 explains why that ordering is load-bearing.

§ 07

Handling changes mid-flight

Three mechanisms, and then the matrix that falls out of them.

(a) Changes apply at checkpoints, not instantly

> **Checkpoint** — a defined safe moment in a worker's execution. Here: just before each model call, and just after a tool call returns.

A worker has an inbox of pending changes and reads it only at checkpoints. So nothing is torn apart in the middle of an API call, and the system stays predictable enough to test.

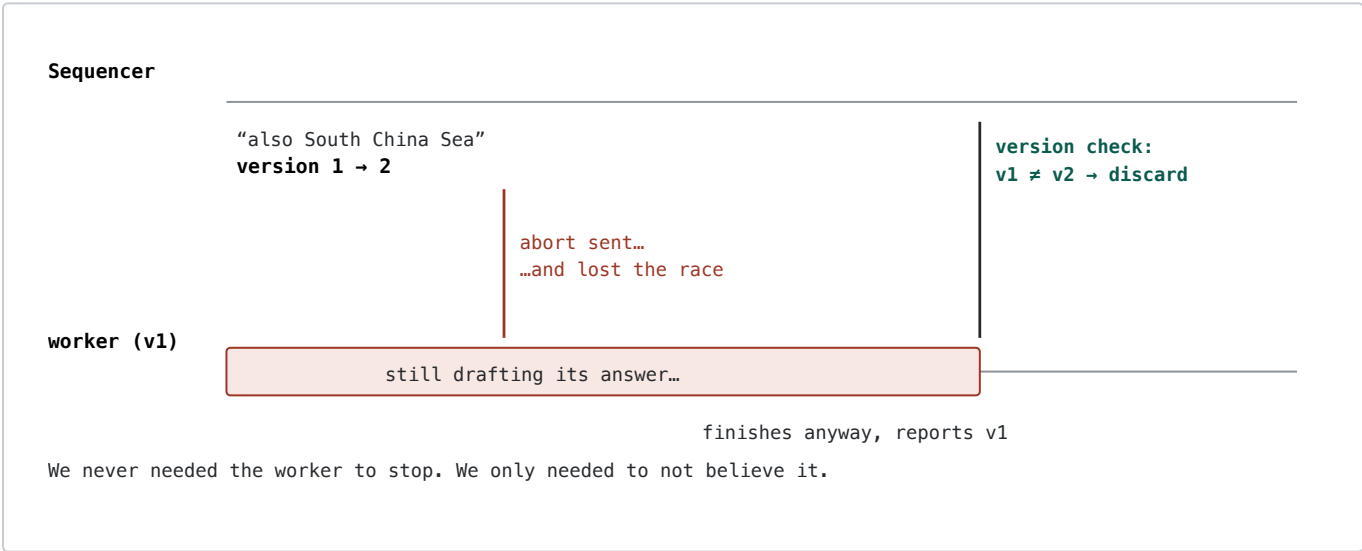
Cost: a change may wait up to one tool call before it takes effect. That is stated as a tradeoff, not hidden. The alternative — interrupting a worker at an arbitrary instruction — buys a few seconds and costs the ability to reason about, or test, what state the worker was in when you stopped it.

(b) Version checking makes reliable cancellation unnecessary

The Sequencer ignores any result whose version is not the current one.

That sounds small and is the most important mechanism here. Reliably *stopping* work that is already running, across a network, is a genuinely hard problem — your stop signal races the work finishing, and sometimes loses. *Ignoring* work that came back out of date is trivial and cannot fail.

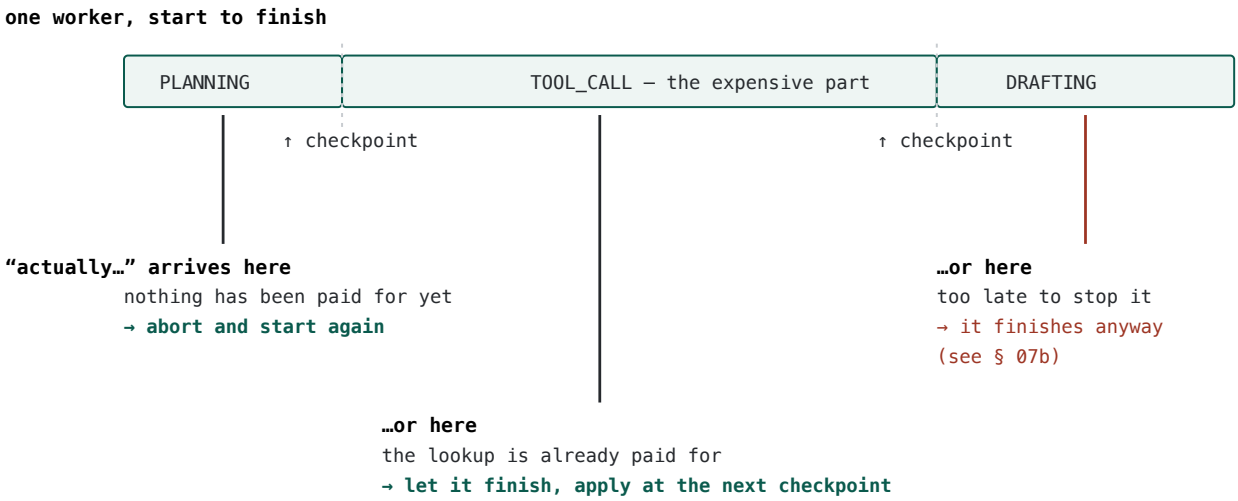
So we still try to stop superseded workers, to save money. But correctness never depends on that succeeding. This is not hypothetical: a stop signal sent while a worker is already writing its final answer *does* lose the race, the worker finishes, and it reports a result for a version that no longer exists. Version checking is what makes that harmless — and § 10 shows it failing on command when the check is turned off.



This is not hypothetical. The simulator models the race faithfully, and `--no-version-gate` turns the check off so you can watch the out-of-date answer reach the chat instead.

(c) Preserve work in proportion to what it cost

Tool calls are expensive in both time and tokens, and are frequently still valid after a change. A draft is cheap to rewrite. So a change that **preserves the premise** rewrites the answer over the tool results it already has, and one that **replaces the premise** throws them away and starts over.



The same message, three different right answers. Which one applies depends only on where the worker had got to — which is why § 07 is a matrix rather than a rule.

The stage × relationship matrix

	QUEUED	PLANNING	TOOL CALL	DRAFTING	DELIVERED
AMEND premise kept	Rewrite in place; nothing lost	New version, start again	Let the tool call finish — already paid for — then apply at the next checkpoint	Rewrite the answer over the tool results we already have	Send an addendum
AMEND premise replaced	Replace in place	Start again	Start again, **throwing away the tool results**	Start again	New answer plus a visible “this replaces my earlier answer”
CANCEL	Drop, acknowledge	Stop, acknowledge	Let the tool call settle, then drop	Stop, acknowledge	Acknowledge *“already answered”* — no retraction
DUPLICATE	Attach the second asker; one answer, sent to both				Send the existing answer to the new asker

The Delivered column is the one most designs omit. A change can always arrive after the answer has already shipped. “Too late” is a real state with real required behaviour, not an error.

§ 08

Edge cases handled on purpose

These are the places a plausible-looking implementation quietly breaks. Each is handled explicitly and each has a test. **Four of the six were found by the fuzzer described in § 10**, not by being clever in advance.

THE TRAP	RESOLUTION
A cancelled parent with a waiting child	Ben asks “and the second biggest?”, then cancels the first question. The follow-up is now waiting on an answer that will never come. When a request ends without ever answering, everything waiting on it is dropped with an explicit notice naming the question that died. **Silence is the one unacceptable outcome.**
Waiting on the answer, not the request’s status	A subtler version of the same thing. If a parent answers, is then amended (which makes it live again), and is then cancelled, its status is DROPPED — but the answer the child needs *already shipped*. Asking “is the parent currently delivered?” strands that child forever; asking “has the answer it needs been delivered?” does not. The gate is written the second way.
Dependency deadlock in the worker pool	If a waiting request could take a worker slot and *then* wait for its parent, a pool full of waiters would deadlock — every slot held by something that cannot proceed, and the thing it is waiting for unable to get a slot. The dependency check therefore happens at **admission**: a request never occupies a slot while waiting on another request.

THE TRAP	RESOLUTION
The ready-but-undelivered window	Between “the worker finished” and “the ordering policy released the answer”, a request has no worker but is not finished either. A change arriving in that window used to do two bad things at once: start a replacement worker while the old slot was already free (**breaking the pool limit**), and leave the now-obsolete answer sitting in the delivery buffer where a later release could still ship it. A change now always discards any undelivered answer, and always goes back through admission rather than starting a worker directly.
A follow-up answering against a different world	“And the second biggest?” builds explicitly on an earlier answer, but if admitted after the snapshot window rolled it would be pinned to newer data — so the two could disagree while one is derived from the other. A dependent request now inherits its parent’s data version and tool results, and an invariant checks it. Found by taking the swarm post’s critique of disposable agents seriously (§ 05).
Two things happening at the same instant	A result arriving and a change arriving can land at the same moment. The Sequencer handles one event at a time, to completion, so there is no interleaving to get wrong — the concrete payoff of keeping the shared-state component free of model calls (§ 05).
Dependency cycles from classifier error	A misclassification could make A wait for B while B waits for A — a loop nobody can ever get out of. Detected when the link is created, and broken by dropping the newer one. The result is a correct answer with weaker ordering, which is the right direction to fail.

THE TRAP	RESOLUTION
Unbounded classifier input	Triage needs to see the live request set, and that set grows. It sees every unfinished request plus the last few answered ones. The live set is bounded; the chat history is not, and is not included.

Deadlock — a standstill where everyone is waiting for someone else and nothing can move. Classically: two people each holding one chopstick, each waiting for the other's.

§ 09

What we measure, and why

Everything below is recorded per scenario, per policy, and printed as a comparison table.

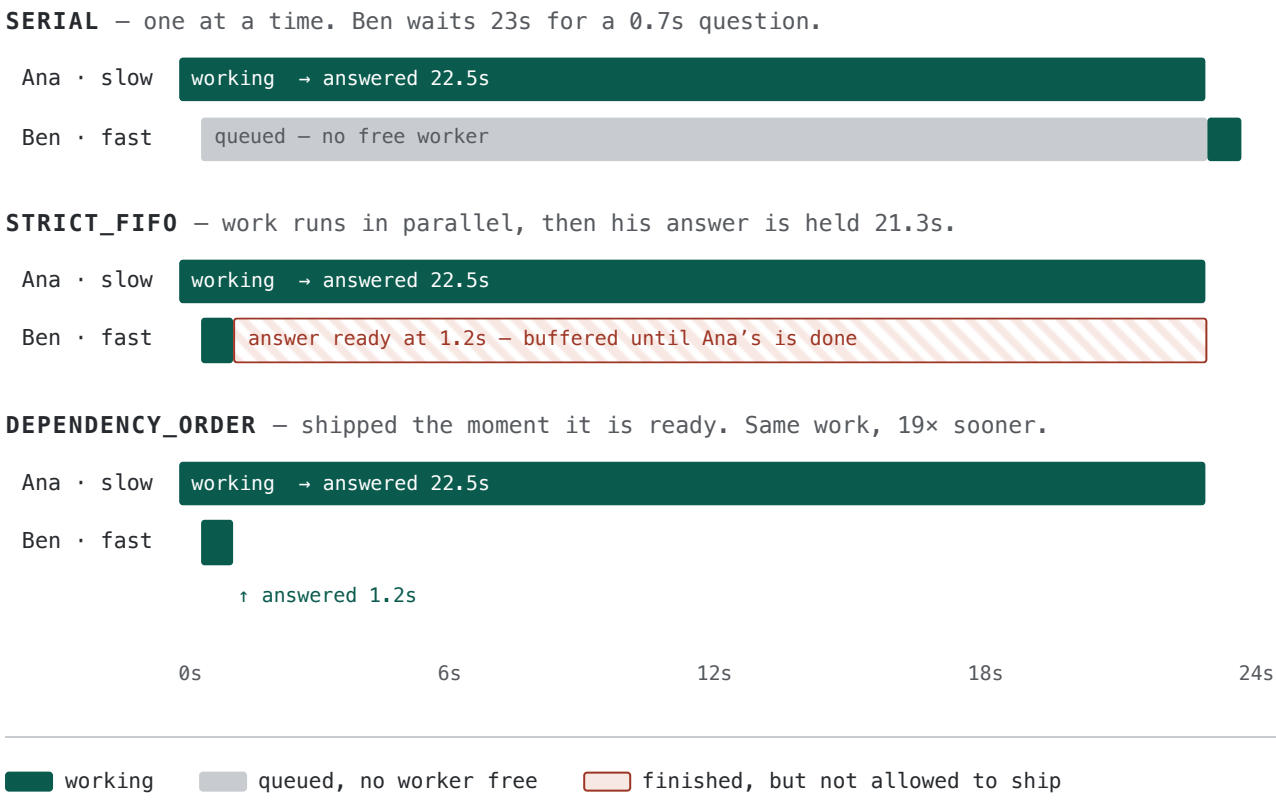
METRIC	WHAT IT TELLS YOU
Time to first / last answer	How long the burst feels. The user-perceived win.
Head-of-line delay	Total time that *finished* answers spent waiting on unfinished ones. Isolates the cost of ordering from the cost of the work.
Wasted tokens	Money spent on versions of requests that were never delivered. **The price of mutability, in dollars.**
Staleness	Answers shipped whose question had already changed. Must be **0** — a correctness check, not a performance number.
Misattribution exposure	Answers shipped with no "re: …" while more than one question was outstanding — the reader had to count positions to know what was being answered.
Sequencer p99	How long the Sequencer spends per event. **Directly tests "serialize the cheap thing."** If this is not tiny, the central claim of § 05 is false.

p99 — the 99th percentile: the value that 99 out of 100 measurements come in under. Used instead of an average because averages hide the occasional very slow case, which is usually the one that matters.

The last three are what make this an engineering argument rather than a rhetorical one. Two of them are designed to be falsifiable: misattribution and Sequencer p99 would both go non-zero if the design were wrong about itself.

Ordering policies, compared head to head

Because § 03 made ordering a policy rather than a structure, these are interchangeable and comparable on identical inputs.



The same two questions, the same amount of work, three policies. Only the striped bar is pure loss: an answer that exists and is being withheld. That is what **head-of-line delay** measures, and it is the entire cost of insisting on arrival order.

POLICY	BEHAVIOUR
SERIAL	One request at a time. **The honest baseline** — the behaviour the brief describes as creating a backlog.

POLICY	BEHAVIOUR
STRICT_FIFO	Work in parallel, but hold each answer until everything that arrived before it is finished. Isolates the cost of literal ordering.
DEPENDENCY_ORDER	Ship as soon as ready, with attribution and acknowledgements; a request waits only for something it genuinely depends on. **Proposed default.**

SERIAL is in the list deliberately. Option 2 of the brief asks for a comparison against "a simple baseline such as SPY alone"; the equivalent discipline here is to measure against the behaviour we claim to be improving on, not only against other clever designs.

§ 10

How we know it is correct

Measuring speed is the easy half. The harder question is whether the thing is *right*, and that needs a different kind of evidence.

Executable invariants

> **Invariant** — something that must be true of the system at all times, no matter what happens. "The pool never runs more workers than it has slots" is an invariant.

The properties this document claims are written as code that checks them, not as prose that asserts them.

INVARIANT	WHAT IT FORBIDS
no_deadlock	Any request still unfinished when everything has settled
completeness	A request the reader was acknowledged for that never produced any visible outcome
no_stale	Shipping an answer whose question had already changed
non_contradiction	The answer text changing without an addendum or replacement notice
causal_consistency	A dependent answer arriving before the answer it depends on
attribution	A policy that promises a requote shipping without one

INVARIANT	WHAT IT FORBIDS
pool_bound	More workers in flight than the pool allows
tokens_accounted	Shipping an answer that no worker ever actually computed
snapshot_coherence	Answering against a different data version than the one the request was pinned to

Measuring the classifier

Everything else tests the *scheduler*. The classifier needs a different instrument, because its failures are judgements rather than bugs.

`python -m chatsched eval` runs 38 labelled messages — each against a realistic live request set, 14 of them marked genuinely ambiguous — through real Haiku, and reports per-relationship precision and recall, a confusion matrix, and the thing § 04 actually claims: **which direction the errors go**.

RELATIONSHIP	PRECISION	RECALL	SUPPORT
NEW	1.00	0.81	32
AMEND	0.86	1.00	36
CANCEL	1.00	0.95	20
DUPLICATE	1.00	0.75	8
REFERENCE	0.89	1.00	16
IGNORE	0.97	0.95	40

One four-pass run: 143/152 = 94% agreement, 94 of 96 clear-cut. A second identical run scored 95%. Quoting a single pass would overstate the precision available here — the spread is the finding, not noise around it.

Two measurement details matter more than the headline number:

- **It repeats.** The classifier is not deterministic — identical passes score between 95% and 97%. A single pass is an anecdote, so `--repeats` runs the whole set several times and reports the spread. The variance is not a footnote: it was 92–97% before § 04's chat context, which is how the missing input was found.
- **It separates the hard cases.** A good score on “never mind the Taiwan one” would hide a bad one on “no”. Clear-cut and ambiguous cases are reported apart.

> The labels are my judgement, which is the honest limitation: this measures agreement with me, not ground truth. What it found is in § 04, and it *changed the design* — which is the point of building it rather than asserting the thresholds were fine.

Randomised testing

> **Property-based testing** (or *fuzzing*) — instead of writing example cases by hand, you generate thousands of random inputs and assert that your invariants hold on every one. Hand-written tests cover the failures you thought of; this covers the ones you didn't.

Because the Sequencer is a pure function (§ 05) and the simulator is deterministic, a randomly generated chat can be replayed in well under a millisecond. So the test suite generates random message streams — arbitrary mixes of new questions, amendments, cancels, duplicates and references, at arbitrary times, under all three policies and every pool size — and asserts all nine invariants on each. Fifteen thousand such chats replay in about forty seconds, without a single network call — `HYPOTHESIS_PROFILE=soak pytest tests/test_fuzz.py`.

This found four real bugs that the hand-written tests did not, all listed in § 08: the orphaned follow-up, the delivered-then-cancelled parent, the pool overrun in the ready window, and the stale answer surviving that same window. It also falsified two of the invariants as too crude — which was worth as much, because it forced them to state the property precisely rather than approximately.

A correctness property you can break on command

> Claiming "staleness is always zero" is weak. Being able to make it non-zero on demand is strong. The stop-signal race in § 07b is modelled faithfully: a worker already writing its answer finishes regardless. Running the same scenario with version checking disabled ships the obsolete answer — and the invariant checker catches it, along with a second violation noticing that the delivered answer was never computed by any worker. Turn the check back on and both return to zero.

That is a flag, not a thought experiment:

```
$ python -m chatsched run --scenario late_amend                    # stale: 0, invariants hold
$ python -m chatsched run --scenario late_amend --no-version-gate  # stale: 1, two violations
```

The second command prints the violations and the stale answer that caused them. Both runs are also asserted as tests, so the demonstration cannot quietly stop working: one test requires the gate to hold, and the other requires it to fail when switched off. A safety mechanism nobody has ever watched fail is indistinguishable from one that does nothing.

Crash recovery, demonstrated the same way

The same standard applies to § 14's claim that recovery falls out of the Sequencer being a pure fold. Asserting it is cheap; showing it is not:

```
$ python -m chatsched run --scenario transcript --journal runs/t.jsonl
$ python -m chatsched recover runs/t.jsonl --crash-after 6

  15 events in runs/t.jsonl
  recovered 3 requests under DEPENDENCY_ORDER, pool 3

    r1  v2  DELIVERED  Ana   What's my exposure to a Taiwan escalation?
    r2  v1  DELIVERED  Ben   Is NVDA still my biggest position?
    r3  v1  DELIVERED  Ben   And the second biggest?

  crash after event 6 of 15, recover, carry on:
    ✓ identical to the run that never crashed
```

The test suite is stricter than the demo. For every scenario it kills the run after **every** event in turn — not one chosen point — recovers from the log, feeds it the events it had not yet seen, and asserts the final state is identical to the run that never died. The fuzzer then does the same to randomly generated chats at a random crash point.

That last part matters more than it looks. Recovery that produces a *plausible* state is worse than no recovery, because nobody notices it is wrong.

Scenario suite

SCENARIO	WHAT IT TARGETS
1	The § 01 transcript — everything at once; the headline demo
2	Burst of six simultaneous questions — parallelism and pool saturation
3	Slow Q1, fast Q2 — head-of-line blocking, in isolation
4	Amendment lands mid-tool-call — a change at the worst possible moment
5	Cancel after the draft is written — wasted-spend accounting
6	Amendment during the final draft — the stop-signal race; the version-check demo
7	Premise replaced mid-flight, then again after delivery — throwing work away, and the replacement notice
8	Two members, same words — deduplication must not cross a portfolio boundary
9	A genuinely busy chat — every relationship at once, including the ones that cost nothing

Tools are mocked with **controllable latency** — the simulator decides exactly when a tool call returns. This is a deliberate upgrade over real tools rather than a shortcut: reproducing "the amendment arrives *during* the tool call" requires controlling precisely that, and a real API makes the race unreproducible.

Workers also have a **live mode** that makes real `claude-opus-5` calls against a portfolio fixture, with real `claude-haiku-4-5` triage — so the demo shows the architecture doing the actual job, not just moving tokens around. Scenario replay uses mocks for determinism; live mode is for the demonstration.

§ 11

Assumptions

Stated as assumptions because several are load-bearing, and a reader should be able to reject one and immediately know what breaks.

ASSUMPTION	IF IT IS FALSE
Chats are independent	The Sequencer stops being per-chat, and the no-collisions property goes with it.

ASSUMPTION	IF IT IS FALSE
`Who the chat is for is explicit, not inferred`	The brief says “a group chat” without saying whose portfolio is being discussed, and the answer changes the design. Shared portfolio (a household, an investment club, a member and their advisor) → two people asking the same question really is one piece of work. Separate portfolios → identical words are different questions, and one answer for both would disclose one member’s positions to another. Rather than guess, each request carries the member it is about; `DUPLICATE` requires a match. **Left implicit, the deduplication in § 04 would be a data-disclosure bug rather than an optimisation.**

ASSUMPTION	IF IT IS FALSE
Within a chat, the only real ordering requirement is dependency	The thesis of § 03, and the most important assumption here. If a compliance rule demands literal arrival order, STRICT_FIFO already exists as a policy and we pay the blocking knowingly.
Tool calls cannot be interrupted once started	Conservative. If they can, cancellation gets cheaper and the Tool call column of § 07 simplifies.
Model calls can be interrupted	True over HTTP — though § 07b exists precisely because interruption can still lose the race.
The classifier is fallible	Its errors are made asymmetric by design (§ 04).
Answer volume per chat is low enough that acknowledgements are not spam	If false, acknowledgements collapse into a single edited status message.
Single process	The event log is the boundary where splitting across machines would go.

§ 12

What is built, and what is only designed

The cut line is itself a design decision, so it is stated rather than hidden.

Not all of it was planned. The fuzzer and the classifier eval each found things that *changed* the design rather than confirming it — four scheduling bugs, and the discovery that § 04's confidence mechanism does not work — and the list of what is built grew as a result. The cut line itself still holds: `MERGE` / `SPLIT`, `BOUNDED_REORDER` and cross-chat scheduling are not built, for the reasons given below.

Built

- Request register, lifecycle, dependency links, and the Sequencer as a pure function
- Worker pool, change inboxes, checkpoint handling, version checking
- Triage over the six relationships of § 04, with the confidence asymmetry, and every decision auditable
- Delivery: attribution, acknowledgements, addenda, replacement notices
- Three policies, nine scenarios, the full metrics table
- Nine executable invariants, a randomised fuzzer, and the version-check demonstration
- Live mode: real Haiku triage and real Opus workers against a portfolio fixture

Designed, not built

- **MERGE / SPLIT relationships.** The highest implementation cost in the § 07 matrix — folding two running workers together, or spawning a second out of one — for the rarest real cases, and the two categories a small classifier is least reliable at telling apart. The matrix slot is designed; the code is not written. Adding them is additive, not structural.
- **BOUNDED_REORDER(k) policy.** Ship out of order, but never more than *k* places ahead of the oldest outstanding question. A real idea; three policies already prove the argument.
- **Fair sharing across chats.** One very busy chat can currently monopolise the worker pool. A per-chat cap is the obvious knob; real fair-share scheduling is not built.
- **Real Slack or Discord integration.** Authentication and webhooks are not what the brief grades. The replay harness demonstrates the same system better.
- **A database behind the event log.** The append-only log is enough to rebuild any chat (§ 14). Indexing, retention and compaction are real work, and none of it changes a scheduling decision.
- **Retries, backoff, rate limiting.** Real, orthogonal, and well understood.

§ 13

Known failure modes

Listed because a design document that claims no weaknesses is not describing a real system.

- **Genuinely ambiguous messages.** *"Actually, Q3"* may be an amendment or a new question; a human would ask which. The § 04 bias mitigates this; it does not solve it. Across repeated eval runs the residual errors are not scattered — they land on the same messages every time. Two of them — *"is that still running?"* and *"drop it"* — were fixed by giving the classifier the recent chat (§ 04). What remains is *"what about bonds?"* (labelled `NEW`, read as `AMEND`), which is genuinely ambiguous, and Cara's repeat of Ana's question (`DUPLICATE` → `NEW`), which is under-merging — the recoverable direction.
- **Questions about the system are not a category.** *"Is that still running?"*, *"did you get that?"*, *"how long will this take?"* are not portfolio questions and not chit-chat, but the taxonomy has no slot for them — with § 04's chat context they now land on `IGNORE` reliably, which is the safe reading — no worker is dispatched to answer a question about the scheduler — but the member still gets nothing back. The Sequencer holds the answer and needs no model to produce it. A `STATUS` relationship would close this; it is not built, and it is the change I would make next to § 04.
- **The ready-but-undelivered window is handled, not eliminated.** A change landing between "answer ready" and "answer shipped" now correctly discards the pending answer (§ 08), but the reader may still see an acknowledgement followed by a different answer than they were expecting. Correct, mildly disorienting.

- **Cross-chat pool starvation.** One very busy chat can monopolise the workers. Per-chat caps exist; fair sharing does not (§ 12).
- **Dependency cycles from classifier error.** Detected and broken (§ 08), at the cost of weaker ordering for that pair.
- **Unbounded chat history.** The live request set is bounded and the classifier's input is bounded by construction (§ 08), but a very long-lived chat will eventually need its transcript summarised.
- **Member identity is trusted, not verified.** The system keeps one member's answers away from another, but takes the member on the inbound message at face value. Real access control belongs at the chat boundary, not here.
- **The classifier is the weakest link.** Everything downstream depends on it relating messages to requests correctly, and that is a genuinely hard judgement. The asymmetric thresholds make its errors recoverable rather than destructive, and the decision log makes every call auditable — but neither makes it right.

§ 14

How far from production is this?

Honestly: it is a working prototype and a design artifact, not a system you could put in front of members on Monday. The scheduling core is in good shape; almost everything around it is deliberately absent. Being specific about which is which seems more useful than a verdict.

CONCERN	STATE	WHAT IS MISSING
Scheduling correctness	ready	Nine invariants, 15,000 randomised chats per soak run, every relationship and policy exercised.
Behaviour under change	ready	The § 07 matrix is implemented and tested, including the cases that arrive too late.
Classification quality	measured	95% agreement across repeated runs — roughly 1 message in 20 is misread. Tolerable because the errors are shaped to be recoverable, not because they are rare: `IGNORE` and `CANCEL` are exact, and what remains is under-merging. The residual misses are listed in § 13.
Surviving a restart	missing	Every event is written to an append-only log before it is handled, and replaying the log rebuilds the chat. Tested by killing the run after *every* event in every scenario, recovering, and asserting the final state is identical.

CONCERN	STATE	WHAT IS MISSING
Sustained load	missing	No backpressure. If messages arrive faster than the pool drains, the queue grows without limit and nobody is told.
Many chats at once	missing	One Sequencer per chat, but nothing supervises them. A single busy chat can take the whole worker pool.
Who may see what	partial	Members' questions and answers are kept apart, but the member on an inbound message is taken at face value. Real access control belongs at the chat boundary.
Cost control	partial	Waste is measured but not acted on. No budget ceiling, and no debouncing of someone typing three corrections in four seconds.
Operability	missing	No tracing, no metrics export, no alerting. The event log is the only forensic surface.
Real chat integration	missing	No Slack or Discord transport. Deliberate — see § 12.

The pattern is deliberate. Everything marked ready is something the brief asked about; everything missing is infrastructure that is well understood and would have crowded out the part being judged.

What I would build next, in order

- **Backpressure.** The one missing piece that turns a busy day into an outage rather than a slowdown.
- **Debouncing changes.** The system already measures wasted spend; it should act on it instead of only reporting it.
- **Fair sharing across chats.** Per-chat caps are the obvious knob; proper scheduling is not written.
- **Exactly-once delivery across a restart.** Recovery rebuilds *state* correctly, but the commands the fold regenerates — including answers that already reached the chat — are discarded rather than reconciled. A worker that was mid-flight when the process died is simply lost work. Getting this right needs the delivery journal consulted on replay, which is a day's work and was not the part being judged.
- **Streaming delivery** — and an answer to the question it raises: what does “this replaces my earlier answer” mean when half of that answer has already been read? I do not know, and that is the most interesting open question here.

§ 15

Questions you might reasonably ask

Each of these is a decision that could have gone the other way, with the reason it did not.

Why is the Sequencer a pure function? That looks like dogma.

It buys three specific things. The claim in § 05 — that the shared-state component does no inference — stops being a promise and becomes something the shape of the code enforces. The races in § 08 cannot occur, because one event is handled start to finish. And it is the only reason the fuzzer exists: a function with no I/O can be driven tens of thousands of times per second.

The cost is that something else has to carry out the commands, and that seam is where two real bugs appeared — live mode silently ignoring fields the Sequencer sent it. The invariants now check for exactly that.

Isn’t “ordered” just FIFO? You redefined the requirement to make it easier.

That is the fair challenge, and the answer is that FIFO is one way to buy four properties, not the property itself. Ask what a reader needs — to know what an answer answers, what is still outstanding, which answers are live, and that a follow-up arrives after what it follows — and only the last involves order at all.

It is also falsifiable rather than rhetorical. `STRICT_FIFO` is implemented, so if literal order is genuinely required you switch one policy and pay 21 seconds of head-of-line delay knowingly. The argument is that you should not pay it by accident.

Why version numbers rather than cancelling properly?

Because reliably stopping distributed work is hard and ignoring out-of-date work is trivial. The stop signal races the work finishing and sometimes loses — which is not hypothetical, it is modelled and demonstrable. We still try to stop superseded workers, to save money, but correctness never depends on it landing.

Five relationships. Why not three, or nine?

Each one earns its place by producing behaviour the others cannot: `REFERENCE` creates the only dependency edge in the system, `DUPLICATE` is the only one that avoids paying twice, `IGNORE` is the only one that costs nothing at all. Widening, narrowing and replacing a question collapse into one class with one boolean, because the only thing that differs is whether the research survives.

`MERGE` and `SPLIT` are designed and not built: they carry the matrix's highest implementation cost for its rarest cases, and they are the two a small classifier is least able to tell apart.

You measured your own mitigation and it failed. Why keep it?

Because the thresholds still catch a genuinely low-confidence tail and cost almost nothing. What changed is what the document claims: they are no longer described as the mechanism that makes errors asymmetric, because the data says they are not.

The replacement is a different signal entirely — whether the message names what it refers to. "Never mind the Taiwan one" does; a bare "no" does not, and might not even be addressed to the agent. That distinction is clean where confidence is not.

Why mock the tools? Real ones would be more convincing.

The opposite, for the thing being tested. Reproducing "the amendment arrives *during* the tool call" requires controlling exactly when that call returns; a real API makes the race unreproducible and the scenario becomes a coin flip.

So replay is mocked and deterministic, and a live mode runs the same Sequencer against real Haiku and Opus so the architecture can be seen doing the actual job. Both exist because they answer different questions.

What is the weakest part?

The classifier, and it is not close. Everything downstream depends on it relating messages to requests correctly, and at 95% agreement roughly one message in twenty is misread. That is why the errors are shaped rather than merely counted, why destructive readings ask instead of acting, and why every decision it makes is logged with its confidence and reasoning.

The second weakest is that it is all one process with no backpressure. A busy enough day degrades into an unbounded queue rather than a graceful slowdown.

What surprised you?

That the fuzzer found four real bugs the hand-written tests did not, and that two of them lived in the window the design document had already named as a known weakness — identified in prose, unhandled in code.

And that my own agent had head-of-line blocking in its ingest loop: a stalled classifier call would block every later message, in a system built to remove exactly that.

What would you change about the approach, not the code?

I would build the eval first. Measuring the classifier changed a design decision; writing the thresholds down from intuition and defending them in prose for several hours did not. The same is probably true of the ordering policies — the comparison table settled arguments the design document could only assert.

Glossary

TERM	MEANING
Agent	A program that answers questions using a language model, calling tools to look things up along the way.
Tool call	The agent fetching external data — holdings, prices. The slow, expensive step.

TERM	MEANING
Token	The unit language models are billed in, roughly a piece of a word. “Wasted tokens” are ones we paid for and threw away.
Message	One line somebody typed in the chat.
Request	One piece of work the agent owes an answer for. Several messages can relate to one request.
Parallel	Several requests being worked on at the same time.
Worker	One running job, answering one request.
Pool / slot	The bounded set of workers allowed to run at once; a request waits for a free slot.
FIFO	“First in, first out” — strict arrival order.
Head-of-line blocking	A slow item at the front of a queue delaying everything behind it.
Deadlock	A standstill where everything is waiting on something else and nothing can proceed.
Race	Two things happening at once where the outcome depends on which lands first.
Version	A counter on each request, incremented whenever its question changes. Used to recognise and ignore out-of-date results.
Stale	A result computed against a question that has since changed.
Checkpoint	A defined safe moment in a worker’s run where it reads pending changes.
Classifier	A model that sorts each message into one of the six relationships of § 04.
Confidence	The classifier’s own estimate, 0 to 1, of whether it is right.
Pure function	Code that takes input and returns output without touching files, network, or the clock. Same input, same output, every time.

TERM	MEANING
Deterministic	Produces exactly the same result every run — which is what makes replay and fuzzing possible.
Invariant	Something that must be true of the system at all times.
Property-based testing / fuzzing	Generating thousands of random inputs and asserting the invariants hold on all of them.
p99	The 99th percentile — the value 99 of 100 measurements fall under.

Castle take-home · Option 3 · Jay Talwar · jt878@cornell.edu
Design written before implementation, so the reasoning can be judged separately from the code.